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By electronic transmission

Mr. Elon Musk
Chief Executive Officer
Tesla, Inc.
Austin, Texas

Re: Exploratory inquiry — a non-programmable boundary and audit layer that leaves the end-to-end network intact

Dear Mr. Musk:

I am the founder of Ghost Autonomy, a pre-silicon research program based in Halifax, Canada, developing what we call an **enforcement-and-accountability layer for bounded autonomy**: a risk-monotone actuation boundary terminating in non-programmable analog hardware, an evidence-admissibility discipline applied *before* sensor fusion, and a constitutionally governed memory in which every change to a system's operating truth is a logged, challengeable event. The program's distinguishing practice is evidentiary: every claim in our corpus carries a graded standing — measured, proven, proposed, or notional — and no claim is permitted to exceed it. This letter observes the same rule.

Tesla's wager is the purest in the field: cameras only, no high-definition maps, an end-to-end network, and a fleet that has accumulated on the order of ten billion supervised FSD miles across more than a million paying users — a data asset no competitor can replicate. We write precisely because of what that architecture concentrates. In an end-to-end driver, every assumption — about visibility, about intervention thresholds, about the effect of each weight update — lives inside the network, where regulators, insurers, and courts cannot read it. The accountability artifacts now demanded of the industry — structured safety cases in California, open federal engineering analyses, data-recorder mandates abroad — are exactly the artifacts an end-to-end architecture does not naturally produce. As unsupervised operation expands from Austin, Dallas, Houston, and Miami toward your announced consumer timeline, the set of assumptions on which that expansion continues is, in our view, the most critical engineering object Tesla owns — and it currently has no external form.

At this point in your program's trajectory, we believe the most consequential engineering object you own is the set of assumptions on which it will continue — and that this set now deserves explicit registration, physical enforcement, and independent audit.

The offer. Ghost is prepared to pivot its development to maximize compatibility with Tesla's architecture — and nothing about the offer asks the network to change. Our admissibility layer is modality-agnostic and runs on a camera-only suite without modification; our guard ECU is conceived as a retrofit at the vehicle's actuation interface, outside the network, constraining nothing in its interior; our conformance instrument consumes fleet trace formats as they exist. The network stays end-to-end; the boundary and the audit trail live beneath and beside it.

What Ghost brings — specifically, and at its true standing.

1. **A measured boundary primitive.** An electrically isolated analog safety veto with ~ 32 ns intervention latency, witnessed on FPGA **[MEASURED]** — deliberately the only latency figure our corpus reports as measured rather than projected.
2. **Proven envelope mathematics.** The risk-monotone actuation algebra in its box case: projection is coordinatewise clipping; enforcement is firmly non-expansive (1-Lipschitz); the two-stage clamp is exact in the feasible case **[PROVEN]**.
3. **A fully specified refusal chain.** Five stages (S0–S4), from trusted numerics through fidelity gating, layered safety filtering, and risk-monotone envelopes to the analog veto, with six acceptance criteria (G1–G6) restated as a conjunctive, fail-closed conformance gate — published as criteria, not achievements **[PROPOSED]**.
4. **An admissibility-before-fusion evidence discipline.** Sensors as witnesses under integrity, alignment, scope, and lineage gates; correlated witnesses counted once; inadmissible evidence excluded rather than averaged in — sensor-agnostic by construction **[PROPOSED]**.
5. **A constitutional memory doctrine.** Append-only canon, anti-silent-drift amendment semantics, verification burden that increases with residence time, and recall as a priced contract (Governed Knowledge Systems trilogy, July 2026) **[NOTIONAL]**.
6. **A self-auditing corpus.** An 89-page consolidated foundational corpus (May 2026); a 94-page technical documentation edition, v3.0 (June 2026); and a 367-page compendium whose provenance ledgers grade every claim — including a published coverage map (68 subcategories: 21 full, 28 high, 9 partial, 10 gap) and a formal proof-debt ledger (P-1–P-7) naming what remains unproven.
7. **Empirical provenance, honestly scoped.** Physics-informed reduced-order kernels on elliptic-integral/AGM primitives with machine-precision convergence **[PROVEN]**, validated at $R^2 \approx 0.78\text{--}0.87$ on 200+ experimental datasets within their demonstrated envelope **[MEASURED]** — in a non-automotive domain, with automotive transfer registered in our own ledger as an open validation obligation.

What this could be worth to your program — specifically.

1. **A structural argument to complement fleet statistics.** A conjunctive conformance gate bounds what a network error can do — the argument that scale alone cannot supply, addressed to the reviews your program currently faces.
2. **Physical fail-closed independence.** A non-programmable veto path beneath the software stack is the strongest available answer to the question every unsupervised-mile skeptic asks — and it requires no concession from the end-to-end interior.
3. **An assumption registry without an architecture change.** Visibility envelopes, intervention thresholds, and update effects registered, enforced at the boundary, and auditable — turning the data flywheel's outputs into checkable claims.
4. **Per-vehicle witness logs for a distributed robotaxi network.** When owner vehicles join the network, signed decision traces give the fleet an accountability substrate that neither the owner nor the company must be trusted to reconstruct after the fact.

What Ghost is not. We have no fabricated silicon, no vehicle deployment, and no fleet data; our measured hardware evidence is presently a single number. We state this plainly because we regard the statement itself as an asset — a partner adopting our layer inherits a provenance

ledger, not a marketing narrative — and because it is precisely why a compatibility-first alignment now, while our interfaces are still being fixed, costs little and shapes much.

Our inquiry is deliberately modest: *is such an alignment of interest?* As a first, zero-obligation step, we would run our conformance instrument against a trace or scenario format of your choosing and return the results with their standings — an artifact your team keeps regardless of what follows. Our two strategy volumes and the published corpus are available on request; deeper documentation under mutual NDA.

Respectfully,

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Evidentiary standings used in this letter — **[MEASURED]** witnessed on hardware; **[PROVEN]** a mathematical result; **[PROPOSED]** specified and defended, not yet validated; **[NOTIONAL]** a research direction, quotable as intent only. It is our house rule that no statement may exceed its standing; we extend the same rule to this letter.